

Every Crown Counts from Above

Bringing SOTA individual tree crown detection to practitioners

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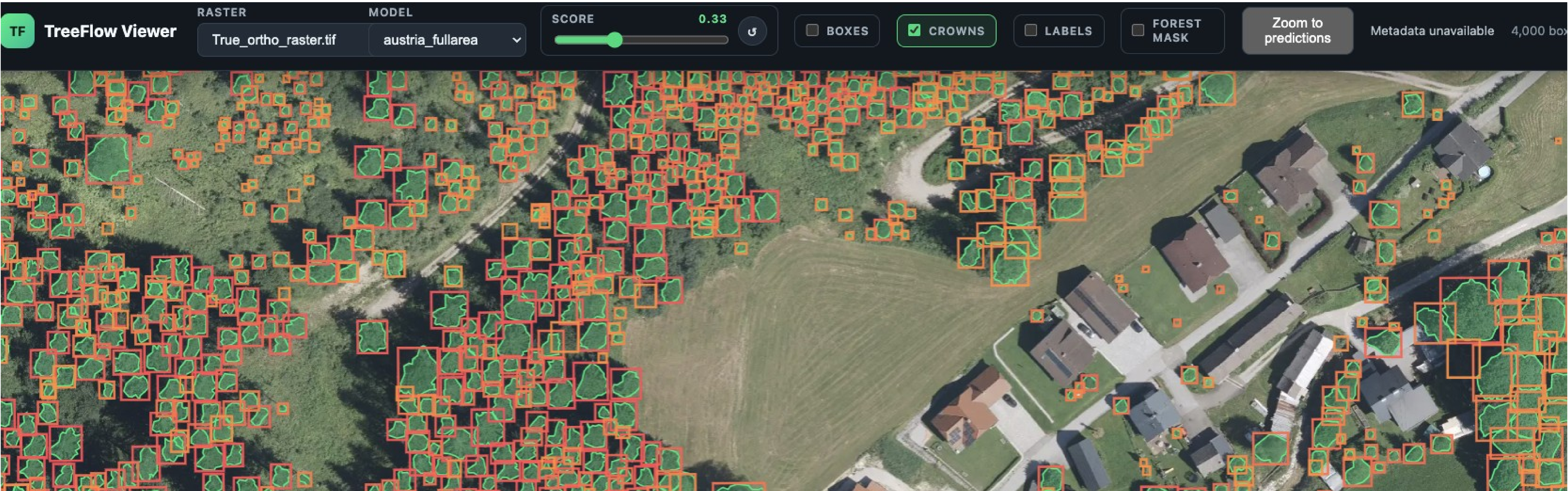
Problem

- Forests are changing faster than we can survey them.
- The **individual tree** is the basic unit of **forest monitoring**, because **mortality**, **species** and growth are defined per tree.
- Institutions across Switzerland already collect **large amounts** of high-resolution imagery.
- Running state-of-the-art crown detection over large areas is still **complicated and expensive**.

Goal

We aim to make detecting individual tree crowns **easy and affordable** for anyone who manages or monitors forests.

Accurate	Matching state of the art on temperate and urban benchmarks.
Efficient	Cheap to train and finetune, and fast over large-area rasters.
Usable	Packaged with tools to inspect predictions and apply them to new data.



Approach

1 Train @ 512 px

Encoder-H (864 M) + detection decoder, supervised on boxes and labels.

RGB tile · 512 px · Quebec⁴

2 Finetune @ 1024 px

Same supervision, finer pixels — smaller crowns become separable.

RGB tile · 1024 px · Quebec⁴

3 Label-free distillation

A small encoder regresses the big encoder's features. No boxes, no labels — then the trained decoder is reattached unchanged.

TEACHER · FROZEN STUDENT · TRAINABLE

Encoder-H · 864 M Encoder-L · 328 M

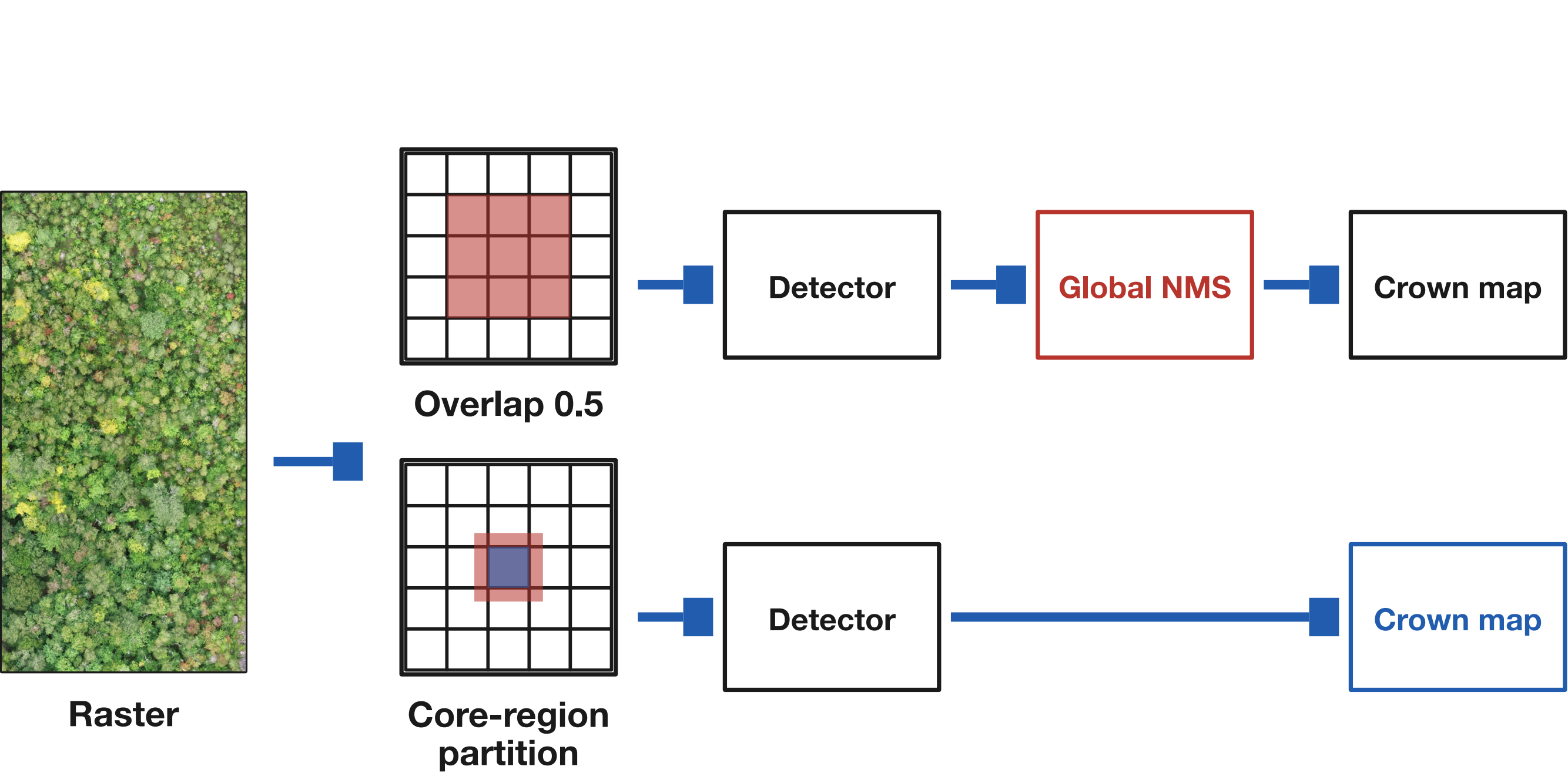
Detection decoder — trained once in step 1, frozen and identical on both sides.

State-of-the-art accuracy at a fraction of the cost

Crown detection on two temperate / urban benchmarks

Model	OAM-TCDB ³ RF1@75	Quebec ⁴ RF1@75	GPU-h RTX 6000	img/s RTX 4090
CanopyRS — current SOTA ¹	52.9	53.0	~43	10.8
ViT-H · @ 512 px	55.0	55.5	5.4	8.2
ViT-H · @ 1024 px	55.5	57.8	8.2	8.2
Distilled ViT-L @ 1024 px	54.9	57.8	13.7	15.2

Core-region partition - Large-area inference



Dense features

PCA of patch features. Pretraining already separates crowns; supervision sharpens the boundaries — a random encoder never gets there, which is why distilling features is enough.

Summary

2.25x fewer windows	+4.1 RF1@75
No NMS	5.2x fewer GPU-hours

Have a look here
for video



[1] Baudchon et al. SelvaBox / CanopyRS. ICLR 2026. · [2] Siméoni et al. DINOv3. arXiv:2508.10104, 2025. · [3] Veitch-Michaelis et al. OAM-TCDB. NeurIPS D&B 2024. · [4] Cloutier et al. QuebecTrees. RSE 2024.